

AnimBP Doctor — 3-Cycle Blueprint

UE5 Editor Plugin: From Python Scripts to Fab Marketplace Product

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Topics Covered:

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8. DevTeam Fluidity Audit — 7 Laws Scoring
9. Fab Submission Strategy
10. Development Timeline & Session Plan

Project Context

Product: AnimBP Doctor v1.0

Price: \$39.99 on Fab Marketplace

Engine: UE 5.4 - 5.7+

Module Type: Editor (stripped from shipping)

Estimated LOC: ~2,000-2,500 C++

Brand: Bifrost Tools

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1. Executive Summary

5 CRITICAL DECISIONS

1. Editor-only module (Type='Editor') — stripped from shipping builds, zero runtime cost
2. Mirror SWHealthChecks pattern — FAnimBPScanResult with MakeError/MakeWarning/MakeInfo factories
3. FScopedTransaction on ALL graph modifications — full undo/redo support
4. IAssetRegistry::GetAssetsByClass(UAnimBlueprint) — no manual path scanning
5. Bifrost dark-theme HTML reports — consistent branding across plugin + docs

Product Positioning

- Gap: ZERO dedicated AnimBP diagnostic tools on Fab marketplace (confirmed via search)
- Closest competitor: Live Blueprint Debugger (generic BP, not AnimBP-specific)
- Our edge: AnimBP-specific checks (T-pose detection, skeleton compat, slot validation)
- Price: \$39.99 (sweet spot for complex code plugins on Fab)

Revenue Projection

Metric	Conservative	Optimistic
Monthly sales	10-15	25-40
Monthly revenue	\$352-527	\$880-1,408
Annual revenue	\$4,224-6,324	\$10,560-16,896
Break-even	Month 1	Month 1

Revenue after Fab's 12% cut (88% to creator).

2. Existing Code Audit

Python Scripts (Porting Baseline)

Script	LOC	What It Does	Port Target
gen_animbp_research.py	~400	AnimBP architecture analysis	Scanner logic
_gen_animgraph_research.py	~800	AnimGraph node enumeration	Scanner checks
gen_animgraph_wiring_pdf.py	~600	Programmatic wiring research	Fixer patterns
setup_sword_and_dummy.py	~280	AssetRegistry + skeleton queries	Scanner discovery

C++ Foundation (Already Built)

System	File	What We Reuse
Health Checks	SWHealthChecks.h	Registry pattern, result struct, Pass/Warn
AnimInstance	SWAnimInstance.h	30+ properties, NativeUpdate pattern
Log System	SWLog.h	Custom log category pattern
Debug Subsystem	SWDebugSubsystem.h	Console command registration

Key Research Already Completed

- Professor_AnimGraph_Programmatic_Control_UE5.pdf — full C++ API audit
- Professor_AnimGraph_Programmatic_Wiring.pdf — node creation patterns
- UAnimGraphNode_* class hierarchy documented (12 node types)
- FBlueprintEditorUtils, FKismetEditorUtilities usage patterns
- Confirmed: UE5 Python API is READ-ONLY for AnimGraph (C++ required)

3. Architecture Design

Class Hierarchy

Class Overview

```
// 4 classes, clean separation of concerns

FAnimBPDoctorModule : IModuleInterface
// Tab registration, menu entries, module lifecycle

UAnimBPScanner : UObject
// Asset discovery + 12 check methods
// Input: content path | Output: TArray<FAnimBPScanResult>

UAnimBPFixer : UObject
// Auto-fix dispatch + graph modification
// All changes wrapped in FScopedTransaction

UAnimBPReport : UObject
// HTML report generation + log output
// Bifrost dark theme, A-F grading

SAnimBPDoctorTab : SCompoundWidget
// Slate UI: toolbar + stats + progress + SListView
```

Data Flow

Data Flow

```
User clicks [Scan All]
|
v
IAssetRegistry::GetAssetsByClass(UAnimBlueprint)
|
v
For each AnimBP: ScanSingleAnimBlueprint()
|-- CheckNullAnimReferences()
|-- CheckBlendWeights()
|-- CheckSkeletonCompatibility()
|-- CheckDefaultSlot()
|-- CheckStateMachineTransitions()
|-- CheckTPoseFallback()
|-- CheckOrphanedNodes()
|-- CheckBlendSpaceConfig()
|-- CheckDeprecatedNodes()
+-- CheckDuplicateSlots()
|
v
TArray<FAnimBPScanResult> -> SListView display
|
v
User clicks [Fix All] -> FScopedTransaction -> Recompile
User clicks [Report] -> HTML file -> Browser
```

4. Scanner — 12 Check Types

#	Check	Severity	Auto-Fix?	What It Catches
1	NullAnimReference	Error	No	Sequence Player with no anim = T-pose
2	BrokenBlendWeight	Warning	YES	Weight outside [0,1] range
3	SkeletonMismatch	Error	No	AnimAsset targets wrong skeleton
4	MissingDefaultSlot	Warning	YES	No Slot node = montages won't play
5	BrokenTransition	Warning	No	Unreachable states in state machine
6	TPoseFallback	Error	No	LayeredBoneBlend BasePose disconnected
7	OrphanedNode	Info	No	Node not reachable from Output Pose
8	InvalidBlendSpace	Warning	No	BlendSpace with 0 or 1 sample
9	MissingNotify	Info	No	AnimNotify refs deleted functions
10	DuplicateSlotName	Warning	No	Same slot name used twice
11	UnusedVariable	Info	No	AnimInstance var never read in graph
12	DeprecatedNode	Warning	No	Node class marked CLASS_Deprecated

Key API Patterns Used

Core Scanner Patterns

```
// Asset Discovery
IAssetRegistry& AR = FModuleManager::LoadModuleChecked
    <FAssetRegistryModule>("AssetRegistry").Get();
AR.GetAssetsByClass(
    UAnimBlueprint::StaticClass()->GetClassPathName(),
    AssetDataList, /*bSearchSubClasses=*/ true);

// AnimGraph Access
UEdGraph* AnimGraph = AnimBP->FunctionGraphs[0];

// Node Type Detection
for (UEdGraphNode* Node : AnimGraph->Nodes)
{
    if (auto* SeqNode = Cast<UAnimGraphNode_SequencePlayer>(Node))
    { /* check sequence reference */ }
}

// Pin Inspection
UEdGraphPin* Pin = Node->FindPin(TEXT("Source"));
bool bConnected = Pin && Pin->LinkedTo.Num() > 0;
```

5. Fixer — Undo-Safe Auto-Fixes

Fix Workflow

- All fixes wrapped in FScopedTransaction — user can Ctrl+Z to undo
- Node::Modify() called before ANY graph change (UE5 transaction requirement)
- FKismetEditorUtilities::CompileBlueprint() after each AnimBP's fixes
- MarkPackageDirty() to ensure save prompt appears

Auto-Fix: Add DefaultSlot Node

DefaultSlot Insertion Pattern

```
FScopedTransaction Txn(LOCTEXT("AddSlot", "Add DefaultSlot"));
AnimGraph->Modify();

// 1. Create Slot node
auto* SlotNode = NewObject<UAnimGraphNode_Slot>(AnimGraph);
SlotNode->Node.SlotName = FName("DefaultSlot");
AnimGraph->AddNode(SlotNode, false, false);
SlotNode->AllocateDefaultPins();

// 2. Find Output Pose input pin
UEdGraphPin* OutputIn = OutputNode->FindPin(EGPD_Input);

// 3. Rewire: Source -> Slot -> Output
UEdGraphPin* OldSource = OutputIn->LinkedTo[0];
OutputIn->BreakLinkTo(OldSource);
OldSource->MakeLinkTo(SlotNode->FindPin("Source"));
SlotNode->FindPin(EGPD_Output)->MakeLinkTo(OutputIn);

FKismetEditorUtilities::CompileBlueprint(AnimBP);
```

Auto-Fix: Blend Weight Clamping

Weight Clamping Pattern

```
for (auto& Weight : RuntimeNode.BlendWeights)
{
    if (Weight < 0.f || Weight > 1.f)
    {
        LBBNode->Modify();
        Weight = FMath::Clamp(Weight, 0.f, 1.f);
        FixedCount++;
    }
}
```

6. Slate UI — Editor Tab

Layout

Tab Layout (ASCII Wireframe)

[illegible]

Key Slate Widgets

Widget	Purpose
SListView<TSharedPtr<FAnimBPScanResult>>	Scrollable results list with column headers
SHeaderRow	Column headers: Severity, AnimBP, Type, Descript
SProgressBar	Scan progress (0-100%), cyan fill color
SButton	Toolbar actions (Scan, Fix, Report, Open, Clear)
STextBlock	Health grade (A-F) with dynamic color binding

7. DevTeam Fluidity Audit — 7 Laws

Law	Score	Evidence
1. Single Spine	5/5	One entry point: ScanAllAnimBlueprints() -> check methods ->
2. Data-Driven	4/5	FAnimBPScanResult USTRUCT is clean; check types are enum-driv
3. Interface Contracts	4/5	FOnScanProgress/FOnScanComplete delegates for UI binding
4. Async by Default	3/5	FScopedSlowTask for progress; true async scan is v1.1 feature
5. Component Composition	5/5	Scanner/Fixer/Report/UI are independent, composable
6. Minimal Coupling	5/5	Scanner has no dependency on Fixer or Report
7. Test Boundaries	3/5	ScanSingleAnimBlueprint() is testable; no automated tests yet

Total Score: 29/35

This is a strong architecture for a v1.0 plugin. The main gaps (async scanning, automated tests) are planned for v1.1.

Anti-Patterns Avoided

- No God class — Scanner, Fixer, Report, UI are separate
- No hardcoded paths — IAssetRegistry for discovery
- No raw UEdGraph manipulation without FScopedTransaction
- No shipping build impact — entire module is Editor-only

8. Fab Submission Strategy

Submission Requirements

- Plugin compiles on UE 5.4, 5.5, 5.6, 5.7 (test each version)
- Custom log category (not LogTemp)
- All LOCTEXT macros with proper namespace
- 128x128 plugin icon (Bifrost-themed)
- 5+ screenshots showing full workflow
- No game-specific code/assets

Pricing Strategy

- \$39.99 — sweet spot for complex code plugins
- Below \$49.99 impulse buy threshold
- Above \$29.99 signals quality (not a toy)
- Bundle discount with Project Doctor (v2): \$59.99 for both

Marketing Plan

- YouTube tutorial: '3 AnimBP Mistakes Every UE5 Dev Makes (And How to Fix Them)'
- UE5 Forum post: 'Free AnimBP health check tool' (lite version or free scan)
- Reddit r/unrealengine: Show off the HTML report output
- Fab Friday participation (50% of monthly Fab revenue comes from events)
- Cross-promote with Project Doctor when launched

9. Development Timeline

Session Plan (4-6 sessions to MVP)

Session	Focus	Exit Criteria
1	Compile + Scan	Plugin loads in editor, scans AnimBPs, returns results
2	All 12 checks	Every check type fires correctly on test AnimBPs
3	Auto-Fix + UI	DefaultSlot insertion works, SListView displays results
4	Report + Polish	HTML report generates, all buttons functional
5	Multi-version test	Compiles on UE 5.4, 5.5, 5.6, 5.7
6	Fab submission	Screenshots, listing text, submission submitted

v1.1 Features (Post-Launch)

- Async scanning with Slate ticker (no UI freeze on large projects)
- AnimNotify audit (find notifies referencing deleted functions)
- Montage slot validation (ensure montages target existing slots)
- Blend space coverage analysis (axis ranges vs sample distribution)
- 'Fix All' with preview dialog (show what will change before applying)
- PDF report export (using Bifrost PDF engine)
- Console command: 'AnimBPDoctor.ScanAll' for CI/CD integration

10. Cycle Evolution Log

Cycle 1 → Cycle 2 → Cycle 3 Improvements

Aspect	Cycle 1	Cycle 2	Cycle 3
Check count	6 checks planned	10 checks designed	12 checks implemented
Auto-fix	Conceptual	2 fix types identified	Full FScopedTransaction code
UI	None	Wireframe sketch	Complete Slate implementation
APIs	General awareness	Specific classes identi	Full code with correct include
Fab strategy	Price only	Competition analysis	Full submission checklist

Key Insights from 3-Cycle Process

- Cycle 1 discovered the SWHealthChecks pattern — this became the architecture foundation
- Cycle 2 web searches confirmed zero AnimBP tools on Fab — strongest market validation
- Cycle 3 existing AnimGraph research PDFs had 90% of the C++ API info needed
- DevTeam audit scored 29/35 — only gaps are async (v1.1) and testing (v1.1)
- Total estimated LOC: ~2,000-2,500 across 10 files

BOTTOM LINE

AnimBP Doctor is a focused, well-scoped product with zero competition.
4-6 sessions from prototype to Fab submission.
Existing Python scripts + C++ health check pattern = 60% of the work is done.
Conservative revenue: \$4,200-6,300/year. Optimistic: \$10,500-17,000/year.
\$39.99 price point. Editor-only. Undo-safe. Bifrost-branded.
Start Session 1: copy plugin folder to UE project, get it compiling.